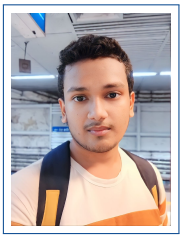
Akshar Nath Gorain

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Portfolio: workforakng.github.io **ORCID:** 0009-0002-0083-3654 **Location:** Kolkata, India



PROFESSIONAL SUMMARY

B.Tech CSE (2024–2028, 3rd Yr) bridging AI research with production engineering — Computer Vision, efficient deep learning, full-stack systems. Co-author: paper accepted and presented at IEEE SPACE 2026, poster at IndiSetu 2026. International champion, COMSYS Hackathon 7.

HONORS & AWARDS

COMSYS Hackathon 7 – International Championship **Škoda Auto University, Czech Republic**
Champion – Rank #1 *June 2026*

- Announced at COMSYS 2026 (7th Intl. Conference on Frontiers in Computing & Systems, Škoda Auto University, Czech Republic).
- Hybrid vision pipeline (YOLOv8, MSER, ConvNeXt/EfficientNet ensemble) for multiscript recognition.

HexaFalls 2, Accio Relevance — JIS University, Team Traction **Kolkata, India**
3rd Prize (USD 200) *July 2026*

Top 5 Best Poster — IndiSetu Kolkata 2026 **Kolkata, India**
AI in Healthcare Summit *July 2026*

Made in JIS, Celeb Edition 2026 — Startup Pitch Competition **Kolkata, India**
3rd Prize *Aug 2026*

Techdarshi Magazine — Edge-AI Chip: AI Without the Cloud **Techdarshi**
Co-author, pp. 26–28 *Sept 2026*

RESEARCH & PUBLICATIONS

Progressive Knowledge Distillation for Military Vehicle Classification on Edge Devices **Co-author**
Techno India University, with Jadavpur University *2026*

- CNN-to-CNN, CNN-to-Transformer and multi-teacher distillation with INT8 quantization and mobile deployment. Accepted and presented at **IEEE SPACE 2026, Bangalore**, July 2026.

Liver Tumor Segmentation from CT Scans using SLIC Superpixel based Swin-UNet **Co-author**
Techno India University *2026*

- SLIC consensus post-processing with fine-tuned Swin-UNet vs MedSAM, SAM2-Tiny, MobileSAM, PicoSAM on LiTS. Poster, IndiSetu 2026 AI in Healthcare Summit, Kolkata.

Multiscript Handwritten Character Segmentation & Recognition **Lead Author**
COMSYS 2026, Škoda Auto Univ. / Jadavpur Univ. *2026*

- Lazy Dataset training, YOLOv8-MSER triple-layer detection, and ConvNeXt-EfficientNet ensemble with script-aware filtering. International Rank #1 (Score: 0.755).

EDUCATION

Degree / Examination	Board	School / Institute Name	Year of Passing	Percentage / CGPA
B.Tech (CSE)	Techno India Univ.	Techno India University, Kolkata	2024–2028	7.69 CGPA
Class XII	CBSE	Guru Gobind Singh Public School, Bokaro	2023	77.83%
Class X	CBSE	Guru Gobind Singh Public School, Bokaro	2021	84.60%

PROFESSIONAL EXPERIENCE

Techno India University — **Kolkata, India**
Undergraduate AI Researcher — *Aug 2024 – Present*

- Efficient deep learning for low-compute deployment and medical imaging pipelines.

CURAA — **Kolkata, India**
Mobile App Prototype & ML Engineer — *Nov 2025 – Nov 2026*

- AI smart pet collar: vocalization → intent, on-device. Cross-platform Flutter architecture, native 60fps hardware rendering, intelligent pet-care service pipelines.

AICTE - EduSkills — **Virtual**
Enterprise AI Engineering Intern — Cohort 19: RAG, Vector Search & MCP — *Oct 2026 – Present*

- In progress — building retrieval-augmented generation pipelines, vector-search indexing and Model Context Protocol (MCP) agent integrations for enterprise workflows.

AICTE - EduSkills (Google for Developers) — **Virtual**
AI-ML Virtual Internship — *Aug 2026 – Oct 2026*

- Completed the 8-week applied AI-ML program (model building, evaluation, deployment fundamentals).

IBM SkillsBuild — **Virtual**
AI Automation & Intelligent Solutions Intern — *June 2026 – July 2026*

- Worked on automation workflows, AI agents and enterprise integrations.
- Certificate ID: BHIBMAC11028

EduSkills Foundation® — **Virtual**
Cybersecurity Intern — *June 2026 – Aug 2026*

- Coursework in network security, threat intelligence and security automation.
- Completed 5 Palo Alto Networks Cybersecurity Fundamentals courses.

EduSkills Foundation® — **Virtual**
Virtual Research Intern — *Dec 2025 – June 2026*

- Ethical Hacking; Generative AI multi-modal pipelines (Google Cloud); Data Engineering, ETL (AWS).

AICTE – Edunet Foundation – Shell India — **Virtual**
AI & Data Analytics Intern (Green Skills) — *Jun 2025 – Jul 2025*

- ML models predicting six major water pollutants from historical environmental monitoring station data.
- Data preprocessing, exploratory analysis, multi-target prediction, model evaluation, inference workflow.
- Certificate ID: STU6829b3f24d2691747563506

PROJECTS

HeartSense — Contactless Vital-Signs Screening — **Kotlin, Jetpack Compose, MediaPipe**
Creator — *github.com/workforakng/heartsense*

- rPPG screening of 5 vitals (heart rate, SpO_2, respiration, HRV, stress) from a 30-second camera scan, 100% on-device with no cloud: AE/AWB-locked CameraX capture, 478-landmark MediaPipe micro-ROIs, POS/CHROM + Butterworth DSP with adaptive Kalman tracking, live 60 FPS BVP waveform, native clinical PDF report.

ShareBite Platform — **Next.js 14, PostgreSQL, NextAuth.js**
Full-Stack Lead Developer — *sharebite-platform.vercel.app*

- Production food-rescue engine (UN SDG 2): real-time surplus ledger, automated category safety windows.

Delta Terminal — **Rust, Python, Flask, AsyncIO, Socket.IO**
Systems & Quant Architect

- Institutional crypto futures & options terminal: BTC/ETH chains with Greeks/IV/OI, Δ-THETA harvesting, Five-Gate regime filter (ADX, IV Rank, Vol Ratio), neural-ensemble inference, fee-aware signal engine, headless CLI for AI agents. Rust core, $\sim$90MB RAM, sub-ms paths.

EdgeRF — RF Drone-Controller Identification for Edge Devices **PyTorch, INT8, Edge AI**
Lead Researcher *2026*

- Paper in preparation (IEEE RadarConf 2027). Achieved 99.89% across 17 RC controllers with a single SNR-agnostic model (prior best 95.62%); 0.59 MB INT8 model with $\sim$2 ms CPU inference. Exposes train/test leakage that inflates published DroneRF results.

VaaniKavach **PyTorch, TFLite, On-device ML**
Co-built — ML & Security

- On-device "voice armor" against AI voice-clone scam calls: distilled anti-spoofing (AASIST $\rightarrow$ TFLite), multilingual coercion-pattern classifier (Bengali/Hindi), family alert mesh, SHA-256 hash-chained evidence ledger.

EdgeECG — Edge ECG Platform **ESP32-S3, Rust, Kotlin/Compose, BLE**
Co-built — Embedded & Edge Systems

- Battery-powered 3-lead ECG screening, 100% on-device analysis: dual AD8232 + ESP32-S3 with Rust firmware, BLE to a Kotlin/Compose app sharing the Rust analysis engine, graded PDF reports.

Legilimens **FastAPI, Next.js, SQLite, Docker**
Co-Creator

- Real-time classroom "confusion radar": confusion flags routed through hybrid BM25 + vector retrieval over an embedded SQLite store to personalized voice explanations, with a professor analytics dashboard.

CERTIFICATIONS

- **Coursera (Oct 2026):** Programming in Python – Meta; Deep Learning Essentials – University of Pennsylvania; Ethical Hacking with Kali Linux – EDUCBA.
- **AWS Academy Graduate - Data Engineering** (Dec 2025) – AWS Training and Certification.
- **Google Cloud Skill Badges:** Develop GenAI Apps with Gemini & Analyze Images with Cloud Vision API.
- **AI and Data Analytics focused on Green Skills** – Completion certificate.
- **AICTE EduSkills** – Data Engineering, Generative AI, Ethical Hacking, AI-ML (Cohort 18).

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, C/C++, Rust, Dart, Solidity, SQL, Bash/Shell, PHP
AI / CV	PyTorch, YOLOv8, ConvNeXt, EfficientNet, Swin Transformer, SLIC Superpixels, SAM family (MedSAM, MobileSAM), TensorFlow Lite, MediaPipe, Gemini API
Cloud & Data	AWS (Data Engineering Certified), Google Cloud (Vision, Functions), Firebase, PostgreSQL (Neon), MySQL, Prisma, ETL Pipelines
Web & Mobile	Next.js 14, FastAPI, Flask, Express.js, Node.js, React Native, Expo, Flutter, Kotlin, Jetpack Compose
Tools	Git, Docker, Linux/Termux Automation, OpenMMLab